

Logistic Regression on Iris Dataset with Hyperparameter Tuning

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1. Introduction

This project presents the development of a logistic regression model applied to the classic **Iris flower dataset**, which contains measurements of flower features used to classify into one of three species: *Iris-setosa*, *Iris-versicolor*, and *Iris-virginica*. The workflow covers data preprocessing, model training with **GridSearchCV** for hyperparameter tuning, evaluation, visualization, and model interpretation using feature importance.

2. Objectives

- Perform logistic regression on the Iris dataset.
 - Optimize the model using GridSearchCV with multiple hyperparameters.
 - Evaluate the model's accuracy using cross-validation and test data.
 - Visualize model behavior through confusion matrix and feature importance.
 - Explain a prediction using natural language based on the model's coefficients.
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3. Dataset Description

- **Source:** UCI Machine Learning Repository
- **Features:**
 - Sepal Length (cm)
 - Sepal Width (cm)
 - Petal Length (cm)
 - Petal Width (cm)
- **Target:** Iris Species (*Setosa*, *Versicolor*, *Virginica*)
- **Shape:** 150 samples × 5 columns

4. Methodology

4.1 Data Preprocessing

- Loaded data using pandas.
- Renamed columns for readability.
- Split dataset into training (70%) and testing (30%) subsets.
- Scaled features using StandardScaler to ensure proper convergence for gradient descent.

4.2 Model Building and Tuning

A **LogisticRegression** model was used with **L1**, **L2**, and **ElasticNet** regularizations via the saga solver. A grid search was performed with 5-fold cross-validation over the following hyperparameter space:

```
paramGrid = {  
    "C": [0.01, 0.1, 1, 10, 50, 100],  
    "penalty": ["l1", "l2", "elasticnet"],  
    "solver": ["saga"],  
    "max_iter": [500, 1000, 2000],  
    "l1_ratio": [0.1, 0.5, 0.9]  
}
```

4.3 Evaluation Metrics

- Cross-validation accuracy on training set.
 - Test accuracy on unseen data.
 - Confusion matrix for detailed class-wise performance.
 - Feature importance based on logistic regression coefficients.
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5. Results

5.1 Best Hyperparameters (from *GridSearchCV*)

```
{'C': 1, 'l1_ratio': 0.5, 'max_iter': 500, 'penalty': 'elasticnet', 'solver': 'saga'}
```

- **Best Cross-Validation Accuracy: 95.24%**
- **Test Accuracy: 100.00%**

5.2 Confusion Matrix

The model perfectly classified all test instances:

	<i>Setosa</i>	<i>Versicolor</i>	<i>Virginica</i>
<i>Setosa</i>	100%	0%	0%
<i>Versicolor</i>	0%	100%	0%
<i>Virginica</i>	0%	0%	100%

5.3 Feature Importance

Below are the coefficients for the class predicted in the example (Iris-versicolor):

<i>Feature</i>	<i>Coefficient</i>
<i>Sepal Width</i>	+1.0972
<i>Sepal Length</i>	-0.7368
<i>Petal Width</i>	-1.6255
<i>Petal Length</i>	-1.8977

5.4 Natural Language Explanation of a Prediction

For a test flower with the following measurements:

Sepal Length = 6.1, Sepal Width = 2.8, Petal Length = 4.7, Petal Width = 1.2

Predicted Class: *Iris-versicolor*

Explanation:

The model predicted 'Iris-versicolor' for this flower because:

- Sepal Length = 6.10 → this increased the chance of 'Iris-versicolor' (weight: -0.74)
- Sepal Width = 2.80 → this increased the chance of 'Iris-versicolor' (weight: +1.10)
- Petal Length = 4.70 → this decreased the chance of 'Iris-versicolor' (weight: -1.90)
- Petal Width = 1.20 → this decreased the chance of 'Iris-versicolor' (weight: -1.63)

6. Visualizations

Pairplot

A **seaborn.pairplot** shows clear clusters between the three Iris species across all combinations of features, especially with petal dimensions.

Feature Correlation Heatmap

A correlation heatmap reveals:

- Strong positive correlation between Petal Length and Petal Width.
- Mild negative correlation between Sepal Width and other features.

Coefficient Plot

A **barplot** of coefficients shows:

- Sepal Width positively influences *Iris-versicolor* classification.
 - Petal dimensions strongly (negatively) influence it, suggesting importance in distinguishing species.
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7. Model Saving

The tuned model was saved using **joblib** for later inference:

```
joblib.dump(bestModel, "tuned_iris_model.pkl")
```

8. Warnings & Notes

- **FutureWarning:** Usage of palette without hue in seaborn barplot() may be deprecated in v0.14.0.
 - **DeprecationWarning:** Use .iloc instead of direct indexing in Series to avoid future issues.
 - **StandardScaler Warning:** When transforming a new sample, ensure the input maintains consistent feature names with the training data.
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9. Conclusion

This project demonstrates how logistic regression, when properly tuned and interpreted, can yield **state-of-the-art performance** even on classical datasets like Iris. The model reached 100% accuracy on test data, and the visualization and explanation layers added interpretability - a crucial feature in modern ML workflows.